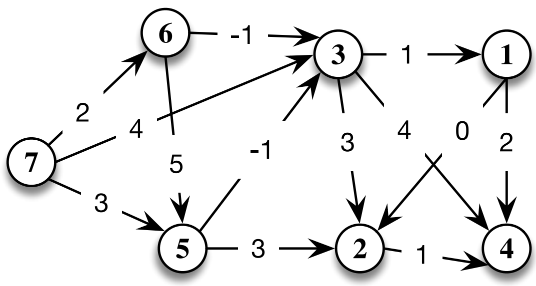
First part : Closed book

Total score: 21 pts. Minimum score to pass: 11 pts. Tentative scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts.

1. Consider the directed graph in the picture. Compute the shortest path tree with root in node 7. Specify what is the best algorithm in the specific case of the type of graphs as that in the picture. Perform a preprocessing of the graph if it is needed. Illustrate the intermediate steps. What is the complexity of the selected algorithm?



2. Consider the following linear programming problem:

$$\begin{array}{lll} \max & 3x_1 - x_2 & \\ & -3x_1 - 2x_2 & \leq -3 \\ & -10x_1 + 6x_2 & \leq 9 \\ & 2x_2 & \leq 7 \\ & 2x_1 + 2x_2 & \leq 11 \end{array}$$

For this problem the trivial solution $x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Why? Formulate the auxiliary problem (AP) that allows us to find a starting feasible solution, if any.

What is the trivial feasible solution$\bar{x}$ for AP? What is the set of active constraints $I(\bar{x})$? Write the conditions for the existence of a growing feasible direction in $\bar{x}$. Is $\xi = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$ a growing feasible direction? What is the value of the moving step and the new point x'? Justify your answers.

3. Consider the following binary knapsack problem:

$$\begin{array}{ll} \max & 12x_1 + 12x_2 + 13x_3 + 6x_4 + 2x_5 + x_6 \\ & 4x_1 + 5x_2 + 6x_3 + 3x_4 + 2x_5 - 5x_6 \leq 5 \\ & x_i \in \{0, 1\}, i = 1, \ldots, 6 \end{array}$$

Solve the problem by applying a branch and bound that uses the linear relaxation as upper bound. Compute an initial feasible solution by applying the greedy algorithm. Report the intermediate steps and the enumeration tree. When possible, use dominance rules to fix variables.

4. Consider the problem $P : \max cx : Ax \leq b, x \in R^n$. Let x' and x'' be two optimal (feasible) solutions, that is $cx' = cx''$, with $x' \neq x''$. Prove that, any convex combination $\bar{x}$ of x' and x'' is optimal.

AMPL questions: 4 points. **in the rear of this paper**

Write the definition in the ampl modelling language of the following:

- a parameter n
- a set `I` of all the integers between 1 and n.
- a non-negative variable x_i with $i \in \{I \cup 0\}$.

Then using the defined variables and parameters write in the AMPL modelling language the following constraint:

$$x_j > \sum_{k=0}^{j-1} x_k \qquad\qquad \forall j \in I$$

$\sum_{k=0}^{j-1}$ means the sum for all indices $k \in \{0, \ldots, j-1\}$

0.1 solution

```
param n;
set I := 1..n;
set I_bar := {I union {0}};
var x{I_bar} >= 0;

s.t. c1{j in I}:
   x[j]  <= sum{k in 0..j-1}  x[k];
```

ANSWER IN THE SPACE BELOW